

		Angular velocity is independent of v , and is directly proportional to B .
10	B	Since the gravitational field strength is uniform near the surface of a planet, $\Delta\phi \propto \Delta r$ Potential difference between 2 points that are 2.5 m apart, $\Delta\phi = \frac{2.5}{10} \times 6.0$ $= 1.5 \text{ J kg}^{-1}$ Work done in raising a mass is the increase in its gravitational potential energy. $\Delta U = m\Delta\phi$ $= 2.0 \times 1.5$ $= 3.0 \text{ J}$
11	D	Speed of the satellite